

SPACEGAME

Design

1. Inspirations
2. Prototypes
3. System Design
4. Game objects

1. Inspirations

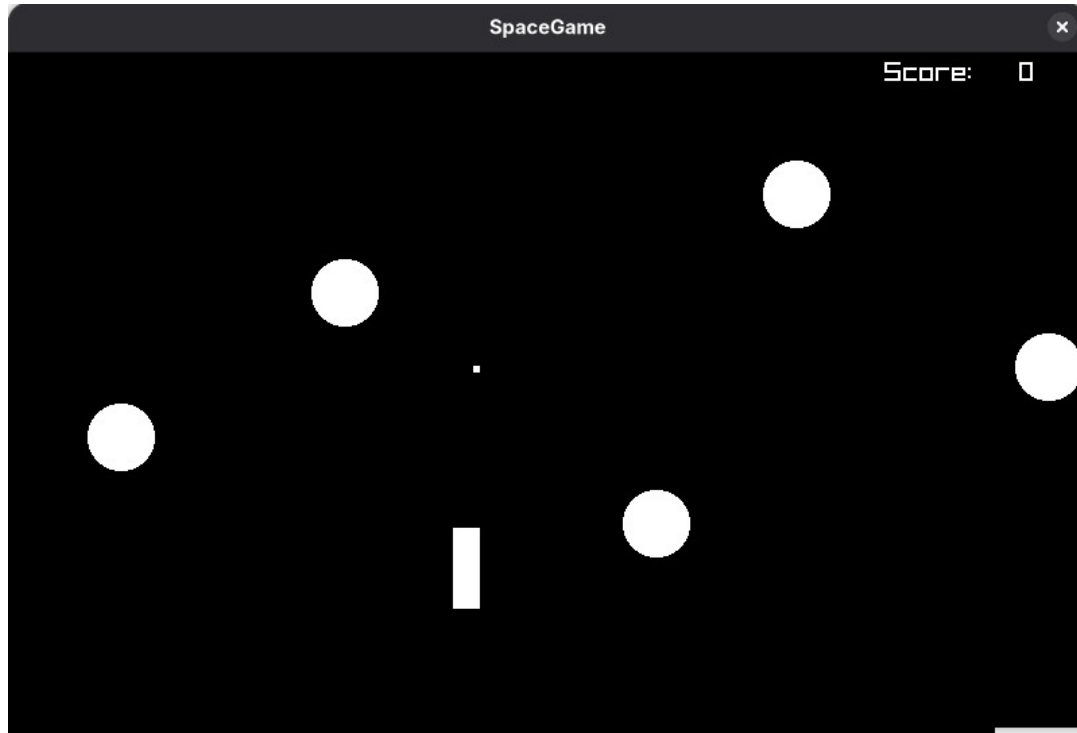


[Asteroids \(video game\) – Wikipedia](#)

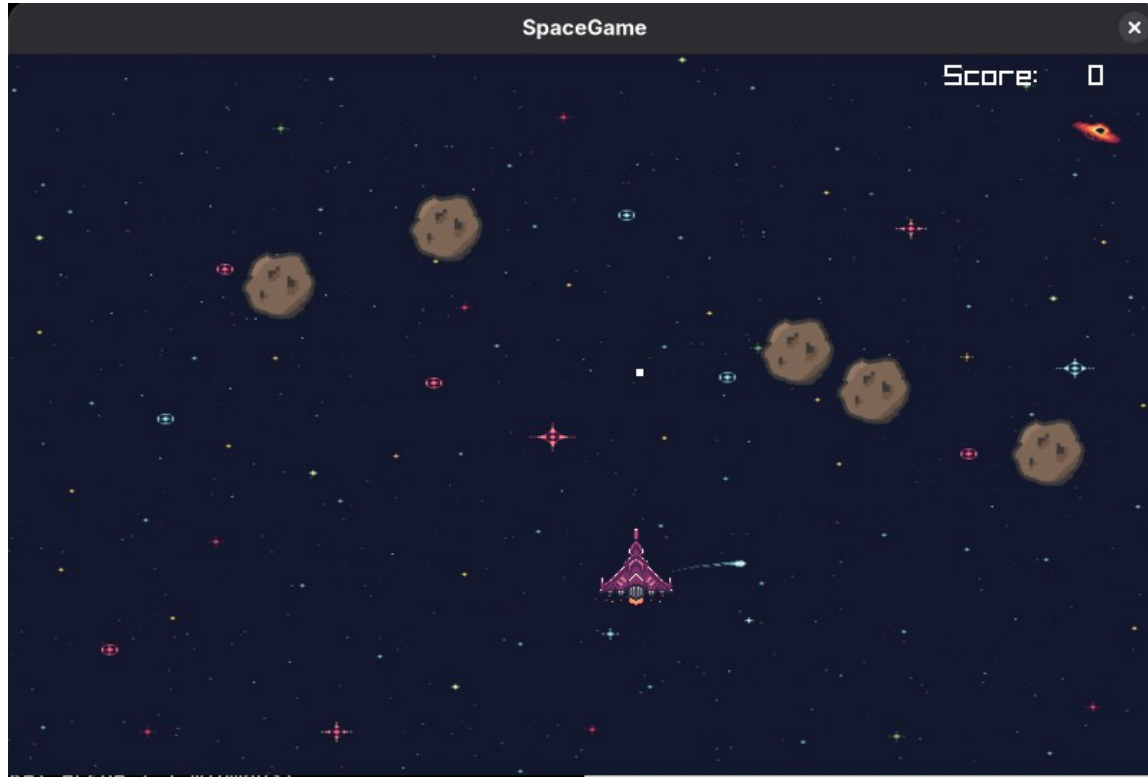


[Third generation of video game consoles – Wikipedia](#)

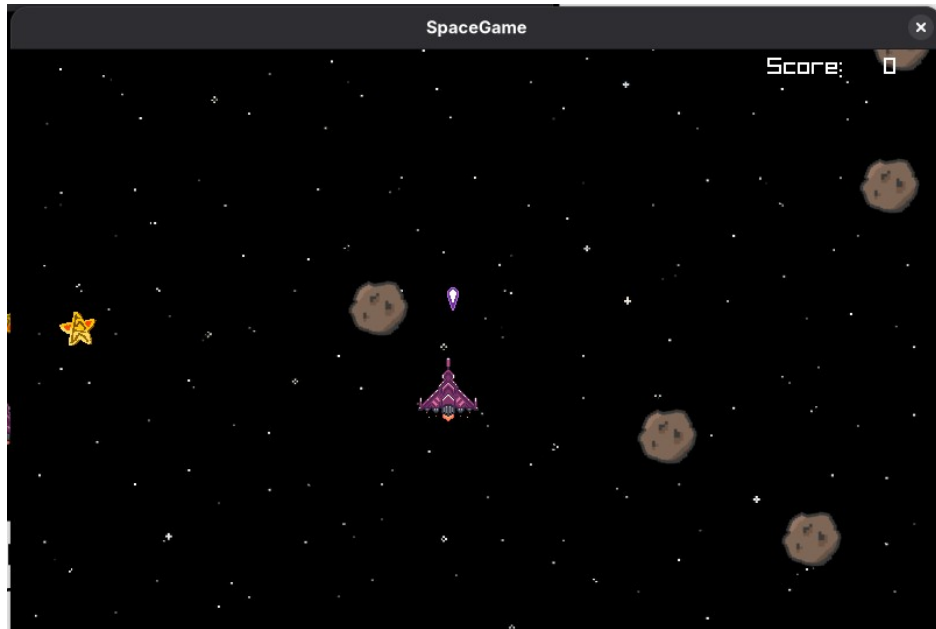
2. Prototypes



1. Early prototype - The image represents the ship as a rectangle, the asteroids as circles, the projectile as a square and the placement of the score counter.



2. Prototype with graphics – static background and images



3. Image showing the reward design and a different background

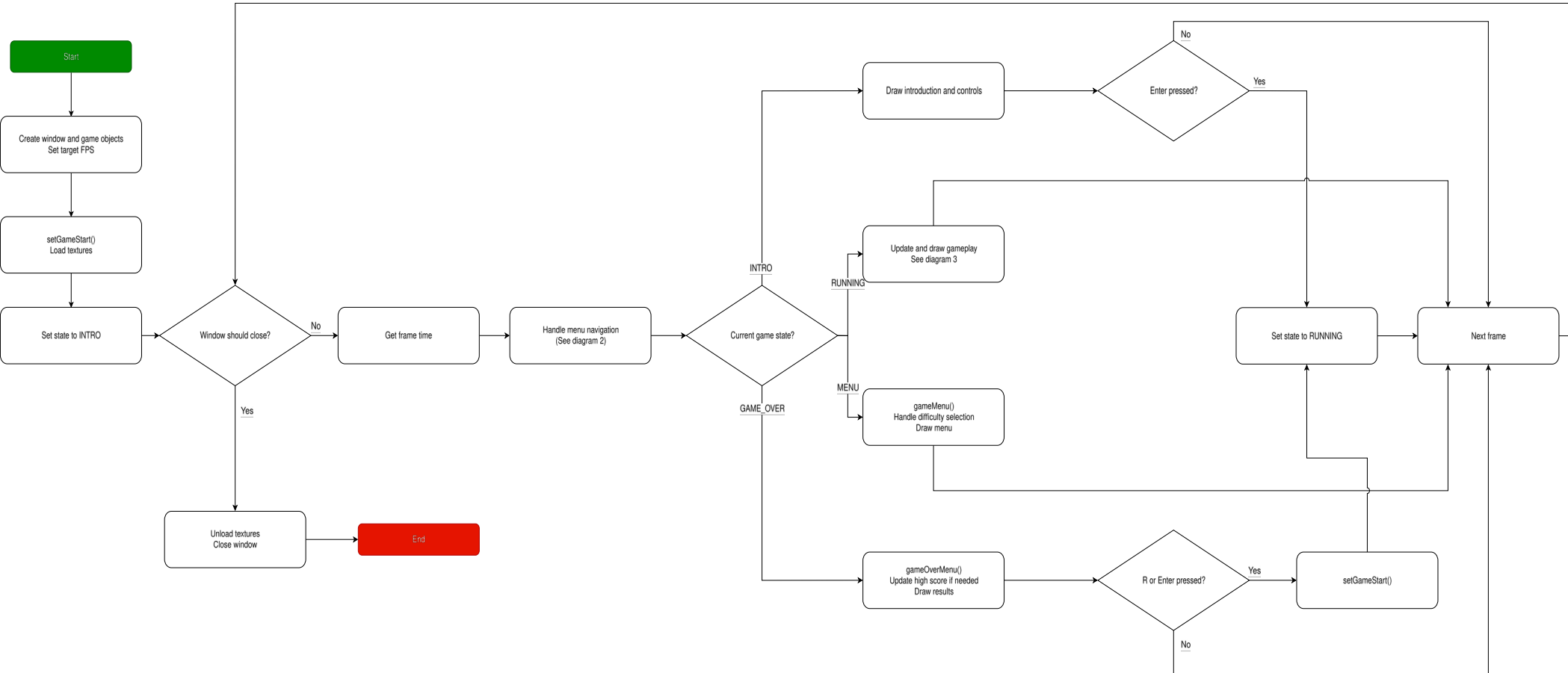


4. Image showing the Game Over menu



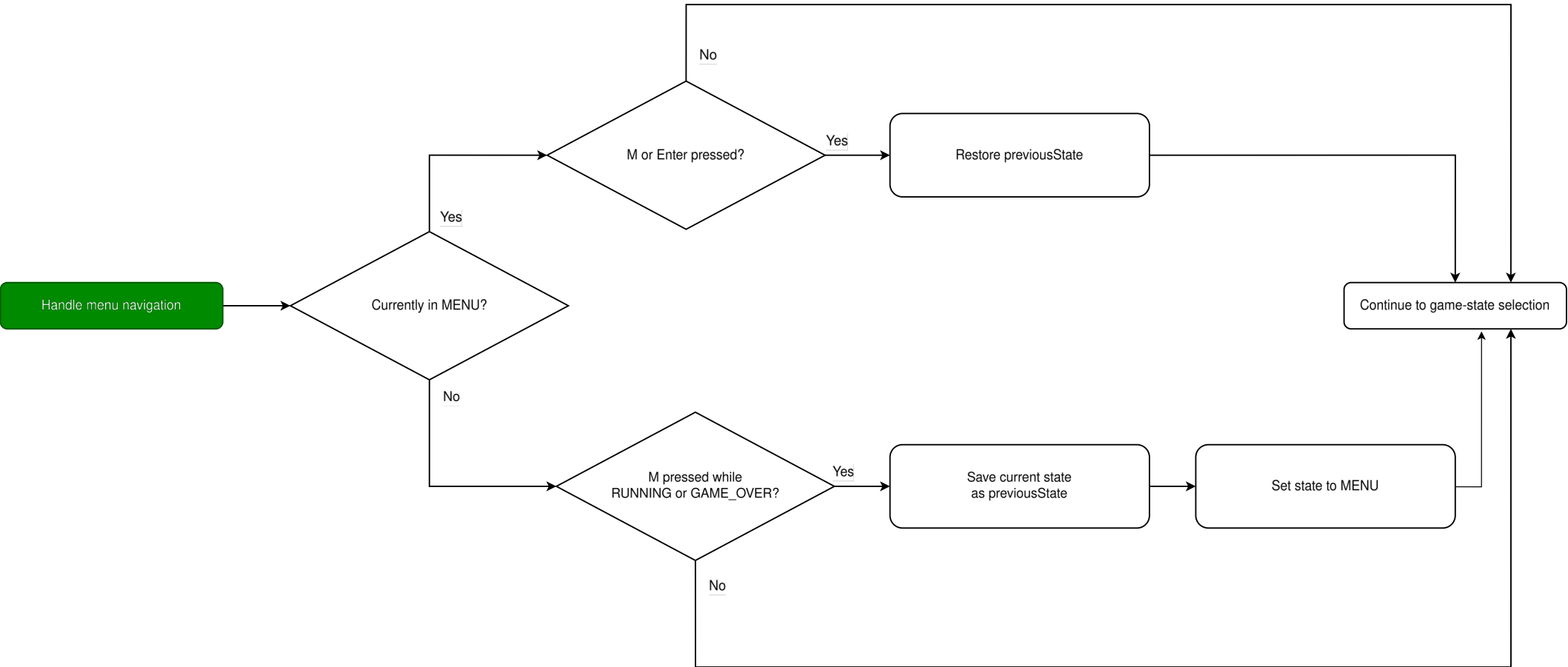
3. System Design

1. Flowchart of the game



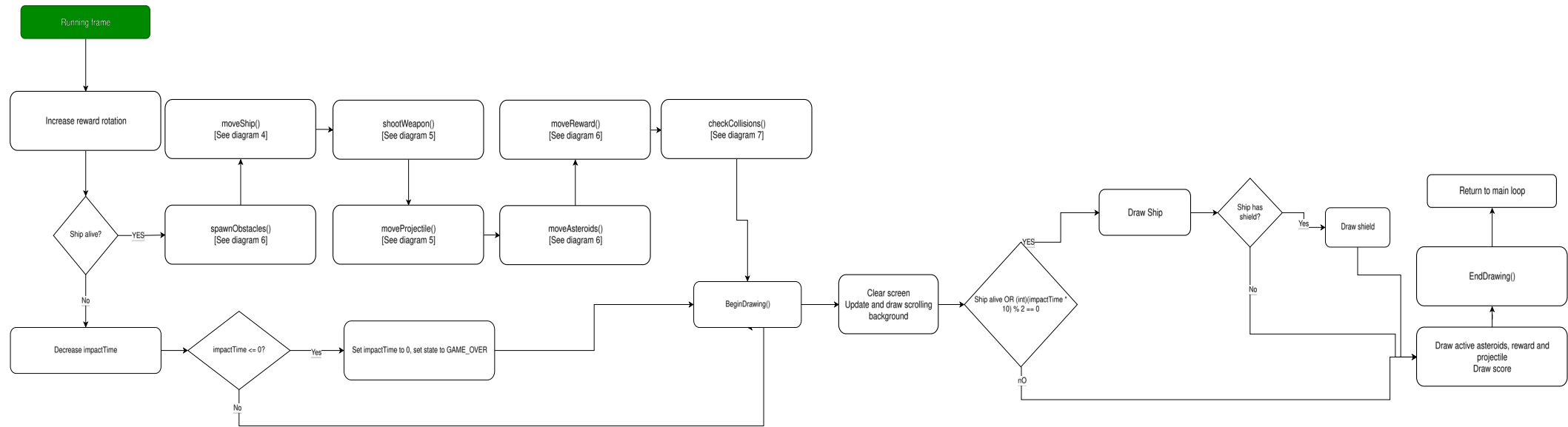
3. System Design

2. Menu navigation flowchart



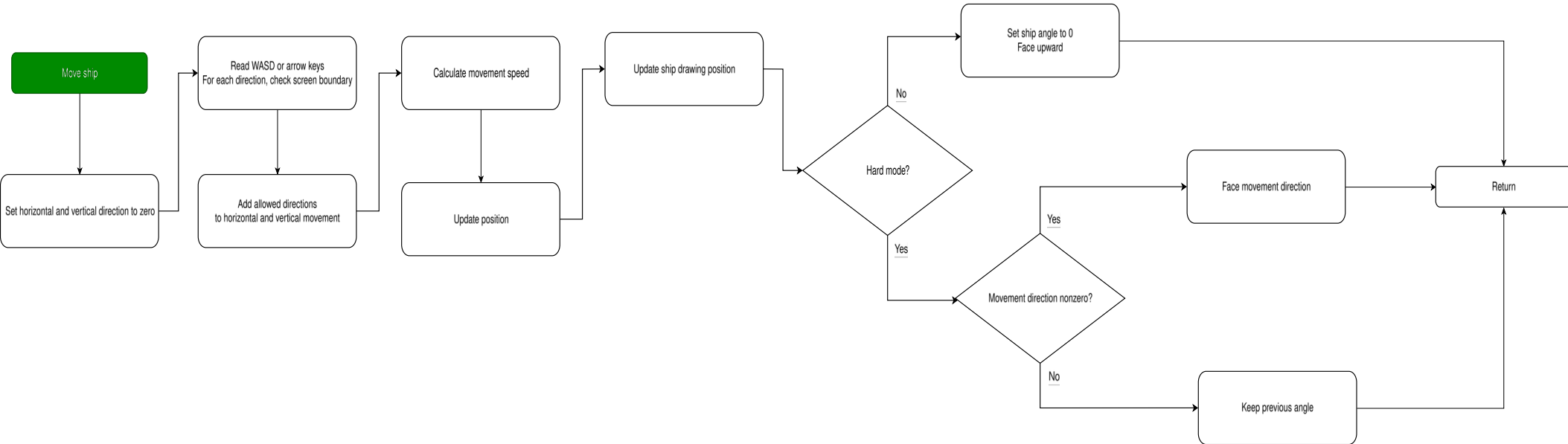
3. System Design

3. GAME_RUNNING flowchart



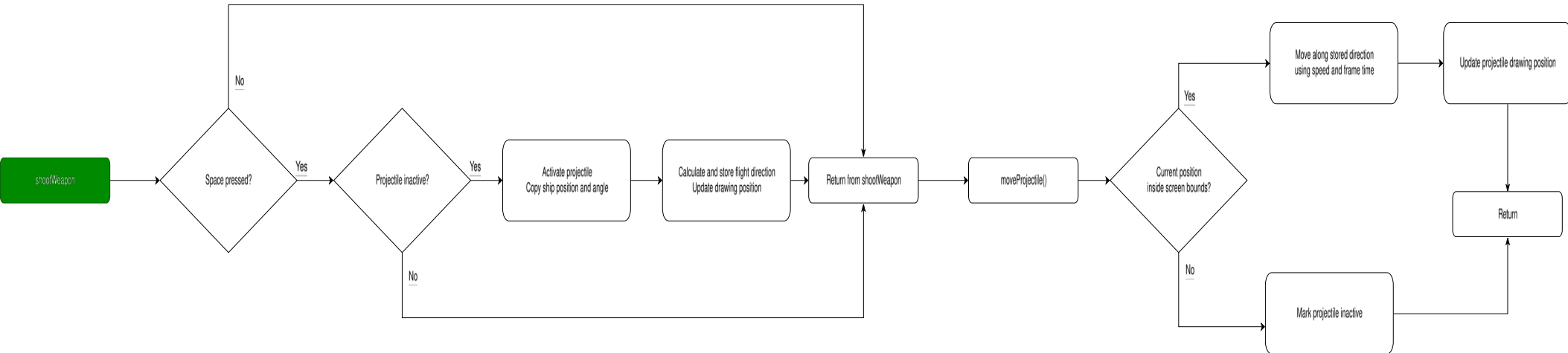
3. System Design

4. Ship movement flowchart



3. System Design

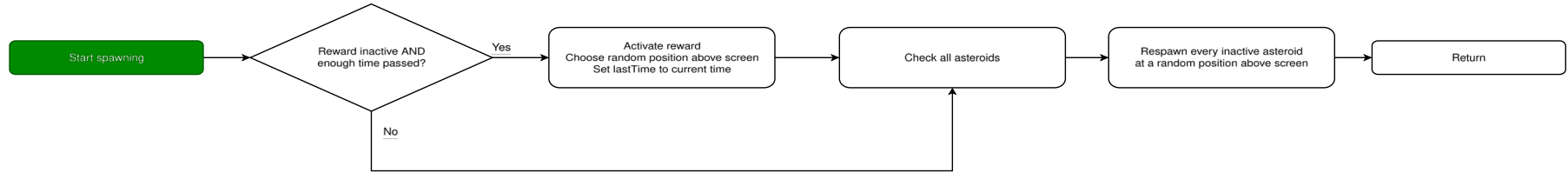
5. Projectile Flowchart



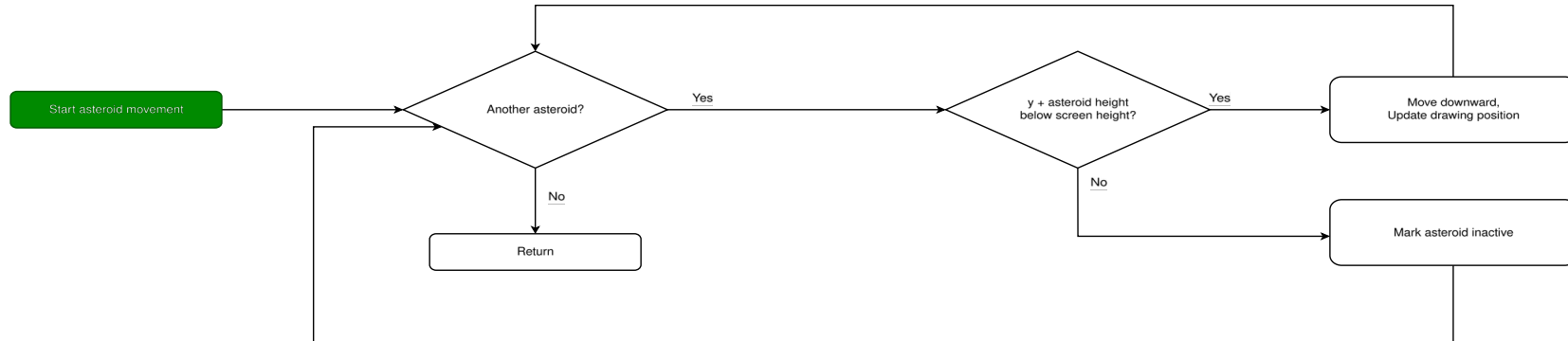
3. System Design

6. Asteroids and Reward flowchart

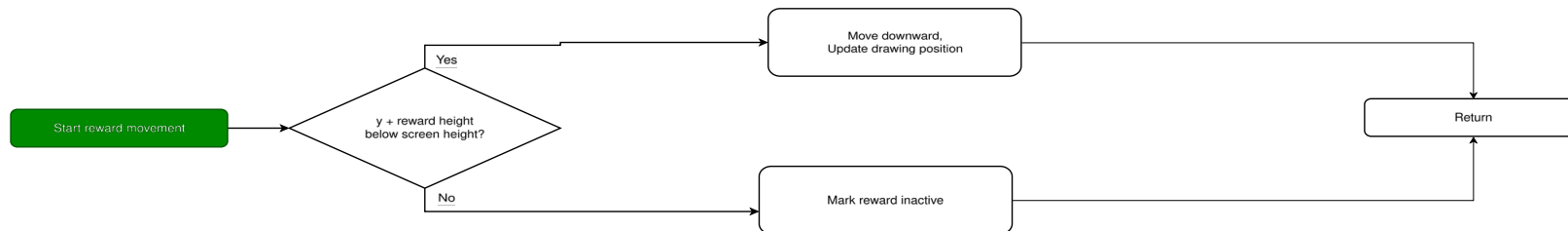
spawnObstacles()



moveAsteroids()

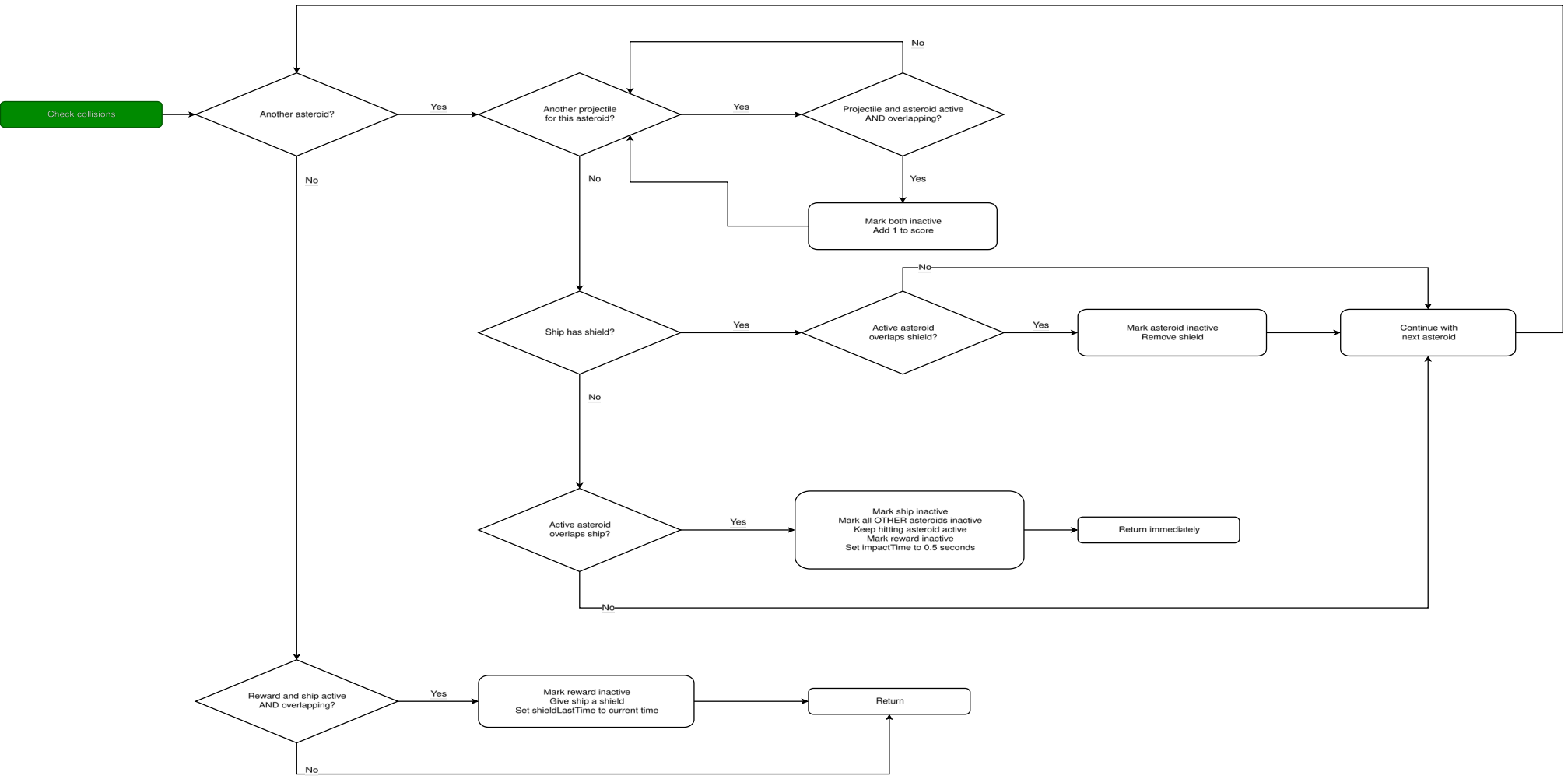


moveReward()



3. System Design

7. Collision checking flowchart



4. Game Objects

Object	Data stored	Behaviour
Player (Player)	Position, radius, speed, alive state, angle, shield state	Moves with WASD/arrows; faces upward in Easy mode or movement direction in Hard mode
Asteroid (Obstacle)	Position, radius, speed, alive state	Moves downwards; respawns when inactive during active gameplay
Projectile (Projectile)	Position, flight direction (dx, dy), alive state, speed, angle	Fires with Space when inactive; keeps the ship's direction at firing
Reward (Obstacle)	Position, radius, speed, alive state	Moves downwards; grants a shield lasting up to 5 seconds or until an asteroid hit